

Bijal Panchal

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SUMMARY

Data Analytics Engineer & BI Professional (M.S. Data Analytics with Distinction) with hands-on experience in data governance, model lifecycle management, and standardized workflow development. I design star schema warehouses, build automated Python ETL pipelines with structured data quality controls, and develop model evaluation and validation frameworks. Skilled in Python, structured and unstructured data handling, feature engineering, and model interpretability techniques, with strong communication translating complex technical findings into clear business narratives. Proficient in Microsoft Office (Word, Excel, PowerPoint). Currently expanding expertise in Databricks. Available immediately for full-time roles.

SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn), SQL, VBA, SQLAlchemy

Modeling & Model Lifecycle: K-Means Clustering, XGBoost, PCA, ARIMA, SARIMA, Holt-Winters, Prophet, Isolation Forest, Feature Engineering, Model Interpretability, Model Evaluation & Validation (Precision, Recall, F1-Score, AUC-ROC), Train-Test Validation

Data Governance & Compliance: Standardized Workflow & Process Development, Data Quality Management, Documentation & Traceability, Root Cause Analysis, Reconciliation

BI/Visualization: Tableau, Power BI (DAX, Power Query), Streamlit, Excel, PowerPoint, Word

Data & Warehousing: Structured & Unstructured Data, MySQL, SQLite, PostgreSQL, Star Schema Design, ETL Pipelines, Dimensional Modelling

DevTools: Git, GitHub, GitHub Actions, Jupyter Notebook, VS Code, pytest

CloudStack: Databricks (developing), Snowflake, dbt, GCP BigQuery, AWS S3/Redshift

WORK EXPERIENCE

POINTS AFRICA

Sep 2025 - Jan 2026

Data Analyst Intern

Toronto, ON (Remote)

- Developed standardized data workflows and governance processes to unify and standardize fragmented datasets from 3 enterprise partners (GLICO, JUMIA, STAR) spanning customer profiles, transactions, loyalty ledgers, and geographic data — cutting data inconsistencies by 30% via automated Python (Pandas, NumPy) and SQL pipelines.
- Achieved 100% transactional field coverage and supported data quality compliance by building iterative ETL workflows with clear documentation and traceability, resolving phone formatting errors, date-time mismatches, and duplicate records at scale.
- Wrote clean, maintainable Python code to architect a star schema data model (4 dimension tables + central Transaction Fact table) with structured governance standards, consolidating multi-partner data into a single source of truth and boosting query performance by 40%.
- Applied feature engineering and RFM segmentation to classify customer loyalty tiers, surfacing high-value clusters across 3 transaction networks to drive targeted partner engagement strategies.
- Delivered 10+ production-ready Tableau dashboards — covering KPIs, partner performance, payment behavior, and geo-regional trends — directly adopted for internal decision-making.
- Translated complex, multi-source structured and unstructured data into clear executive-level narratives, enabling senior stakeholders to make confident, data-driven decisions without needing direct database access.

DMG GROUP OF COMPANIES

Mar 2023 - Sep 2024

Data Analyst

Ahmedabad

- Built and validated a customer segmentation model for IT service clients using Python (Scikit-learn, Pandas), combining K-Means Clustering and XGBoost classification to achieve 87% classification accuracy in identifying high-value client segments.
- Engineered 12+ structured behavioral and transactional features from client usage and contract data, applying feature engineering and PCA-based dimensionality reduction to reduce data noise by 28%.
- Developed standardized reporting workflows and interactive client performance and revenue dashboards in Excel (VBA-automated) and Power BI, consolidating data across 4+ service verticals for real-time tracking of acquisition, utilization, and revenue trends.
- Cleaned, standardized, and governed 4,000+ client records across service and billing datasets using Python and MySQL ETL workflows, improving overall data quality by 30% through structured validation processes.

- Evaluated model performance and interpretability using validation metrics (Precision, Recall, F1-Score, AUC-ROC), documenting findings and presenting actionable insights to management — directly supporting decisions that improved client retention by 18%.

PROJECTS

City Traffic Intelligence Platform | Python, SQLite, Streamlit, Scikit-learn, Prophet — Live

- Engineered a **fully automated real-time data platform** ingesting daily metrics across **5 global cities** from **5 live APIs** (TomTom, OpenAQ, Open-Meteo, Numbeo, City Open Data) — built on modular Python collectors feeding a SQLite star schema warehouse (**9 tables, 4,400+ rows**) via a self-running GitHub Actions pipeline.
- Designed a **weighted City Stress Index scoring engine** (Traffic 35% + Air Quality 25% + Weather 20% + Safety 10% + Cost of Living 10%) powered by **15+ engineered features** — including rolling averages, volatility scores, and heat stress index — using ARIMA, Prophet, Z-Score, Isolation Forest, and K-Means models.
- Deployed a **7-page interactive Streamlit dashboard** (live at city-stress-index.streamlit.app) with real-time stress scores, 30-day forecasts, anomaly detection alerts, and cross-city clustering — validated by a **22-test pytest suite** ensuring pipeline and database integrity.

Retail Sales Forecasting & Demand Planning | Python, SQL, Power BI, Statsmodels, SQLite

- Cleaned and processed **1M+ transactional records** across **43 countries and 4,735 products**, resolving 34K duplicates, 19K cancellations, and 243K missing customer IDs — then engineered a SQL star schema warehouse (1 fact + 4 dimension tables) via a custom Python ETL pipeline (SQLAlchemy).
- Built and benchmarked **SARIMA and Holt-Winters** forecasting models on monthly revenue data, with SARIMA achieving **74.5% accuracy** (MAPE 25.54%) on a chronological 80/20 train-test split — reducing simulated stockout rates by **18%**.
- Delivered a **4-page Power BI dashboard** with dynamic KPI cards, seasonal trend lines, and forecast-vs-actual comparisons (80% confidence intervals) — powering RFM segmentation across 6 customer tiers that identified **843 Champions** (avg. \$5,689 spend) and flagged 409 At-Risk customers as the top re-engagement priority.

EDUCATION

University of Niagara Falls

OCT 2024 - APR 2026

Data Analytics

- Graduated with Distinction, GPA 3.89/4.0 — President's Honour Roll, 100% in Advanced Analytics Internship.

Navrachana University

JUL 2020 - JUN 2024

Computer Science & Engineering

- Coursework in Data Structures, DBMS, AI, Cloud Computing, and Cybersecurity; completed a 12-credit Industry Internship and Major Project.

CERTIFICATIONS & ACTIVITIES

Advanced Power BI: DAX Language, Formulas, and Calculations | Course Completion – Apr 2025

- A targeted Power BI certification covering core skills in data modeling, DAX language, and advanced DAX formulas. Emphasizes practical application through dynamic dashboards and real-world business intelligence scenarios.

Hack For Change | Participant – Apr 2022

- Participated in a 49-hour hackathon organized by Navrachana University in collaboration with NuvYuva. Collaborated with peers to develop innovative solutions aimed at social impact and technology-driven change. Strengthened skills in project development, teamwork, and rapid problem-solving under time constraints._